INCH-POUND
MIL-S-24603A(SH)
5 October 1988
SUPERSEDING
MIL-S-24603(SH)
22 May 1980
(See 6.8)

MILITARY SPECIFICATION

SEWAGE COMMINUTOR, AND ASSOCIATED MOTOR, CONTROLLER, AND DRIVE MECHANISM FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sewage comminutors for Naval surface ship sewage treatment systems and sewage collection, holding, and transfer systems. Included are the requirements for the comminutor motor, motor controller , and motor-comminutor drive mechanism.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.
TT-P-645	- Primer, Paint Zinc Chromate, Alkyd Type.
UU-P-268	- Paper, Kraft, Wrapping.

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4360
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

FEDERAL

PPP-C-850 - Cushioning Material, Polystyrene Expanded, Resilient (For Packaging Uses).
PPP-C-1120 - Cushioning Material, Uncompressed Bound Fiber for Packaging

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-16789 - Pumps (Including Prime Movers and Support Items); Packaging of.
MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-R-20092 - Rubber or Plastic Sheets and Assembled and Molded Shapes, Synthetic Foam or Sponge, Open Cell.
MIL-P-26514 - Polyurethane Foam, Rigid or Flexible, For Packaging.

STANDARDS

FEDERAL

FED-STD-595 - Colors ,

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type 11 - Internally Excited),
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-1472 - Human Engineering Design Criteria For Military Systems, Equipment and Facilities.

2.1.2 Other Government Publication, The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping System.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 463 - Standard Specification for UNS N08020, UNS N08026, and UNS N08024 Alloy Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
AMS 5596 - Alloy Sheet, Strip, and Plate, Corrosion and Heat Resistant 52.5Ni - 19Cr - 3.OMo - 5.l(Cb+Ta) - 0.90Ti - 0.50A1 - 18Fc Consumable Electrode or Vacuum Induction Melted 1750°F (955°C) Solution Heat Treated,

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. , 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational srvnices.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

3.2 Material. Equipment shall be suitable for salt water sewage service in the marine environment, Materials selected shall be corrosion resistant and galvanically compatible with respect to formation of electrolytic couples. Material requirements for the sewage piping systems with which the comminutor shall be compatible shall be as specified in MIL-STD-777.

3.2.1 Cutters. Cutters shall be Inconel 718, cold rolled age-hardened (Rockwell C-40-45) in accordance with AMS 5596.

3.2.2 Cutter spacers. Cutter spacers shall be CRES 304 in accordance with QQ-S-766.

3.2.3 Cutter shaft material. Cutter shaft material shall be CRES 20-CB3 in accordance with ASTM B 463.

3.2.4 Comminutor housing and end-closures. Comminutor housing and end-closures shall be tin bronze in accordance with QQ-C-390, alloy C90300.

3.2.5 Other parts. Other parts which are in contact with the salt water-sewage shall be CRES 304 except by pass screens shall be CRES 316 in accordance with QQ-S-766.

3.2.6 Mechanical seal materials. Opposing seal surfaces shall be of tungsten carbide against carbon, or tungsten carbide against tungsten carbide.

3.2.7 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw material, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 General configuration. Comminutors and the associated motors, controllers, and drive mechanisms shall operate in the shipboard sea water-sewage environment and shall grind or macerate all solids normally found in the ship’s sewage and wastewater drains to particle sizes of 3/8 inch maximum particle diameter and 1 inch maximum length or smaller. The comminutor cutters, drive mechanism, and motor shall withstand impacts and loadings resulting from jamming by hard objects, such as nuts and bolts, without serious damage or performance degradation.

3.3.1 Cutters. Comminutor cutters shall cut, shear, or grind all solids normally found in sewage systems to 3/8 inch maximum particle diameter, and 1 inch maximum length, including towels, rags, paper, glass, hair, bones, organic material, plastics, and thin metal objects.

3.3.2 Screens. Screens or strainers shall not be used to condition the influent to the comminutor. Screens or strainers maybe used to direct flow to the cutters but shall not obstruct the flow path in a manner that will cause blockage or stoppage of flow.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

3.3.3 Case integrity. The comminutor casing housing the cutters or grinders shall be liquid, gas, and odor tight under all normal operating conditions and at internal pressures ranging from minus 1 foot to plus 50 feet of water.

3.3,4 Sewage processing. The comminutor shall process raw, untreated sewage with 1 percent (by weight) solids content at flow rates resulting from a static pressure head of 25 feet of water at the comminutor inlet. Under these conditions , the pressure drop across the comminutor shall not exceed 1 foot of water.

3.3.5 Jamming. Comminutor cutters, cutter drive mechanisms, and drive motor shall sustain a minimum of 30 consecutive jam cycles incurred by heavy metallic objects without the need for repair of the equipment (see 3.9.1, 3.9.4, 4.6.7, 4.6.8, and 4.6.10).

3.3.6 Shock resistance. Comminutor and associated equipment shall withstand high-impact shock as specified in MIL-S-901 for grade B equipment.

3.3.7 Duty cycle. Comminutor and associated equipment shall be capable of both continuous and intermittent start-stop operation under both sustained maximum flow and zero flow conditions.

3.3.8 Mechanical shaft seals. Mechanical shaft seals shall be provided to prevent leakage of sewage from the comminutor (see 3.2.6).

3.4 Physical requirements .

3.4.1 Safety. With the comminutor unit fully assembled and installed for operation, there shall be no safety hazards (electrical, personnel, fire, and so forth) (see 4.5.2).

3,4,2 Standardized flange mounting. For standardization and interchange-ability purposes, the in-line mounting flange-to-flange dimensions of the comminutor casing shall be 10 inches for 4- and 6-inch (nominal influent-effluent pipe size) comminutor and 13-1/2 inches for 8-inch (nominal influent-effluent pipe size) comminutor. Mounting flanges of the comminutor shall be 150-pound flanges in accordance with MIL-F-20042.

3.4.3 Dimensional limits, weight limits, and motor horsepower requirements. Physical size and weight of the comminutor unit (comminutor, motor, and drive mechanism) , exclusive of the motor controller, shall not exceed the limits set forth in table 1. Minimum motor horsepowers for 4-, 6-, and 8-inch (nominal influent-effluent pipe size) comminutor shall be as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

TABLE I. Physical size and weight and horsepower for 4-, 6-, and 8-inch comminutors.

Comminutor (nominal inflow pipe size) (Inches)	Maximum weight (less motor controller) (Pounds)	Maximum unit envelope dimensions (less motor controller) (Inches)	Minimum motor horsepower
4	200	44x23x13	3
6	325	44x23x13	3
8	600	44x23x13	3

3.4.4 Size. Comminutor size denotes the nominal pipe size of the influent and effluent lines. Sizes and quantities provided shall be as specified (see 6.2.1).

3,4.5 Mounting. The comminutor unit shall mount solely by the influent and effluent line flanges without degradation of performance.

3.4.6 Inlet and discharge line diameters. Comminutor inlet and discharge bore size shall be the same as the inside diameter of the connected soil (sewage) piping flanges. Nominal pipe size shall be the same as the comminutor size. Detailed flange dimensions shall be as specified in MIL-F-20042.

3.5 Motor. The comminutor drive motor shall conform co MIL-M-17060.

3.5.1 Motor horsepower. Minimum standard motor horsepower ratings for specified comminutor size shall be as set forth in table I.

3.5.2 Motor characteristics. Unless otherwise specified (see 6.2.1), motor characteristics shall be as follows:

(a) Power and voltage - 3-phase, 440-volt, 60-hertz (Hz)
(b) Enclosure - Totally enclosed fan cooled (TEFC).
(c) Classification - Service C, as specified in MIL-M-17060.
(d) Duty - Continuous.
(e) Bearing type - Ball.
(f) Ambient temperature - 122 degrees Fahrenheit (“F) (50 degrees Celsius (“C).
(g) Temperature rise - 158°F (70°C), as specified in MIL-M-17060.
(h) Design - B, as specified in MIL-M-17060.
(i) Shock - Grade B, as specified in MIL-S-901.
(j) Insulation - B, F, or H, as specified in MIL-M-17060.
(k) Revolutions per minute - Not to exceed 1800.

3.5.3 Other motor characteristics. Requirements for shock resistance shall be as specified in 3.3.6. Motor design and type shall be matched to the comminutor design.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T23:14Z
Check the source to verify that this is the currentversion before use.

3.6 Motor controller. An across-the-line continuous duty motor controller shall be provided in accordance with MIL-C-2212, and shall be compatible with the motor specified in 3.5 and the intended service.

3.6.1 Enclosure. The degree of enclosure for the motor controller provided shall be dripproof (45 degrees).

3.6.2 Control functions, Motor start, stop, and reset functions shall be provided by pushbutton or rotary selector type manual switches located in the controller cover. A switch which will momentarily reverse the direction of rotation of the comminutor to facilitate the freeing of jammed objects shall also be provided,

3.6.3 Protection. The motor controller shall be equipped to provide both overload and low-voltage protection for the motor. The motor controller shall shut down the comminutor in the event of a comminutor jam before damaging stress levels are reached in the comminutor mechanism.

3.6,4 Indicating lights. A pilot light shall be mounted in the motor controller cover to indicate when the comminutor is running. Color of light shall be in accordance with MIL-STD-1472.

3.6.5 Alarm circuit terminal. The controller shall be equipped with terminals that shall actuate an alarm circuit in the event the comminutor motor trips-off the line, or the comminutor stops running.

3.7 Ambient. environment. installation. and operating conditions.

3.7.1 Temperature . Comminutor and associated equipment shall operate at ambient temperatures between 32 and 122°F (O and 50°C) without degradation of performance .

3.7.2 Ship motions. Comminutor shall operate without degradation of performance under ship motions as follows:

(a) Pitching plus or minus 10 degrees with 5 second full period.
(b) Rolling plus or minus 30 degrees with 10 second full period.

3.7.3 Installation and operating orientation. The comminutor shall operate without degradation of performance regardless of attitude or orientation of the installed unit and regardless of changes in attitude or orientation due to list or trim of the ship after installation.

3.7.4 Noise. Comminutor and its associated motors, controller, and drive mechanisms shall meet the requirements of MIL-STD-740, except that airborne noise criteria shall be as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

TABLE II. Acceptable octave band sound pressure levels in dB re: 20 uPa, or dB re: 0,0002 dyne/cm^2.

	Octave band center frequency in Hz								
Frequency	31.5	63	125	250	500	lK	2K	4K	8K
Sound pressure level	81	76	71	63	58	55	54	54	56

3.7.5 Vibration. Comminutors and their associated motors, controllers, and drive mechanisms shall meet the requirements of MIL-STD-167-1, type I, environmental vibration up to and including 50 Hz.

3.8 Service life and operating profile. Design service life of the comminutor shall be at least 10 years with a 50 percent service factor, or 40,000 hours of normal operation. There shall be no limit to the start-stop cycles during the life of the comminutor. Parts subject to wear or deterioration shall have a minimum life of 5,000 actual hours of operation comminuting sewage.

3.9 Reliability. repairability, and maintainability. To the maximum extent practicable, the comminutor shall require a minimum of preventive and corrective maintenance. Potential failure modes shall be anticipated and identified and equipment design shall facilitate the repair actions required under these failure modes . Parts shall be constructed such that they cannot be reassembled incorrectly,

3.9.1 Jamming failure. Failure due to jamming shall not cause significant damage to the comminutor system, The motor controller shall automatically shut down the comminutor in the event of a jam, before damaging stress levels are reached. Also, a single, inexpensive, and easily replaceable component (for example, a shear pin or drive belt) shall fail before damaging stress levels are reached elsewhere in the system in the event the motor controller fails to shut off the comminutor. The total time required to restore the comminutor to operation subsequent to such a failure shall not exceed 2 manhours utilizing on board repair parts and any special tools provided. The comminutor calculations and configuration details will be reviewed prior to manufacture.

3.9.2 Seal-mating surfaces. Comminutor construction shall, to the maximum extent practicable, configure seals and seal-mating surfaces to minimize the risk of damaging or scoring these components during disassembly, maintenance, and assembly of the comminutor.

3.9.3 Special tools. Requirements for special tools for disassembly, maintenance, and assembly of the comminutor shall be minimized. A special tool is defined as any tool not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.9.4 Jam corrective maintenance. Comminutor configuration shall facilitate disassembly and assembly procedures that are necessary to manually restore operation of a jammed comminutor cutter assembly (see 4.6.8). The

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T23:14Z
Check the source to verify that this is the currentversion before use.

procedure for manually removing the cutter assembly, unjamming cutters, and reassembly and restoration of the unit to operating condition shall take less than 1 manhour.

3.10 Identification plates. Comminutor identification plates shall be in accordance with types A, B, C, or D of MIL-P-15024 and MIL-P-15024/5, except that plastic material shall not be used.

3.10,1 Identification plate markings. The plates shall include the following minimum data and information:

(a) Manufacturer's name, model number, and serial number.
(b) Generic name of equipment ("Comminutor"), unit size, and motor horsepower.
(c) Salient design information.
(d) Contract number.
(e) NAVSEA technical manual number.
(f) National stock number (NSN) (where available).
(g) Additional space for DCASMA stamp and for a future NSN if one is not currently available.

3.10.2 Identification plate attachment. Fasteners used to secure the plates to the comminutor shall be corrosion-resistant metallic screws or rivets. The fasteners shall not completely penetrate the comminutor housing or end-covers .

3.11 Painting. Comminutor units shall be painted with a primer coat in accordance with TT-P-645, and one coat of gray paint in accordance with MIL-E-15090 (FED-STD-595 color number 26270).

3.12 Direction of rotation. The direction of rotation of the comminutor shall be etched, stamped, engraved or cast in an appropriate visible location on the comminutor unit. Etched, stamped, engraved or cast markings shall be accordance with requirements of 3.10.

3.13 Lifting eyes. Means shall be provided for lifting the assembled comminutor with a chain-hoist or similar lifting device.

3.14 Welding. Welding shall be in accordance with MIL-STD-278.

3.15 Brazing. Silver brazed joints shall conform to the fabrication and inspection requirements of NAVSEA 0900-LP-001-7000. However, where practical, the use of silver brazed joints shall be avoided.

3.16 Drawings. When specified in the contract or purchase order, drawings shall be prepared (see 6.2.2 and appendix).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as othermise specified in the contract or purchase order, the contractor may use his own or any other

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system, When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination specified in 4.5 and tests specified in 4.6. The comminutor shall be installed in a test facility configured and instrumented to enable all required tests to be conducted. The comminutor installation shall be in accordance with 3.4.5 and the manufacturer's installation instructions and drawings (see 3.16), The same comminutor and associated equipment shall be used for all tests. The serial numbers of the comminutor and associated equipment as well as the contract and order number shall be recorded.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination specified in 4.5.1 and the tests of 4.6.1 and 4.6.3.

4.5 Examination.

4.5.1 Material examination. Equipment, components, and material used by the contractor in the manufacture and assembly of the comminutor units shall be examined upon receipt to assure conformance with specified or approved materials,

4.5.2 Safety examination. The comminutor shall be examined to assure against safety hazards (electrical, personnel, fire, and so forth).

4.5.3 Physical examination. The comminutor shall be examined to determine conformance with the physical requirements of this specification regarding dimension, weight, screen, flanges, and others required by section 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.6 Tests.

4.6.1 Gas tightness. With the comminutor motor running, the internal comminutor casing shall be pressurized with air to approximately 2 feet of water pressure head (gauge), and then the air supply shall be shut off. With pressurized air supply removed, the pressure shall not drop below 50 percent (approximately 1 foot of water) of the original pressure (gauge) reading within a 30-minute period. Contained volume of air used in this pressure test shall not exceed 50 standard cubic feet,

4.6.2 Liquid tightness. With the comminutor running, the internal comminutor casing shall be pressurized with water to a pressure of 50 feet of head. Comminutor shall be operated under these conditions for 3 hours. Repeat test at a pressure head of 10 feet of water. Any perceptible leakage during the test shall be basis for rejection,

4.6.3 No-load motor test. With the comminutor operating under no-load, motor voltage and current draw readings shall be recorded. Checking power draw against the motor characteristics shall verify that no binding or excessive drag is occurring in the comminutor or drive mechanism.

4.6.4 Fineness of grind. With a filter bag or screen placed in the discharge stream from the comminutor, sufficient flush fluid shall inject the following items into the comminutor:

(a) Toothbrush,
(b) Shop rag.
(c) Potato.
(d) Sponge.
(e) String mop strands.
(f) Drinking glass.
(g) Steel wool pad.
(h) Razor blades.
(i) Work gloves
(j) Bottle caps,
(k) Steak bone (at least 6-inches long).

Items shall pass through the comminutor. Fineness of grind shall meet or exceed the requirements of 3.3 and 3.3.1.

4.6.5 Noise test. Noise tests shall be conducted with the comminutors and its associated motors, controllers, and drive mechanisms arranged to duplicate shipboard installation and operation in a normal condition. Noise levels shall be within the limits specified in 3.7.4. When specified in the contract or order, component shop noise test reports shall be prepared (see 6.2.2).

4.6.6 Shock test. Shock tests shall be conducted in accordance with MIL-S-901, with the comminutor not operating, to determine conformance to 3.3.6. When specified in the contract or order, equipment shock test reports shall be prepared (see 6.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

4.6.7 Jam test. With the comminutor operating at no-load, a single washer, nut, or bolt of sufficient size and gauge shall be fed into the cutters to jam (stall) the comminutor. This sequence shall be repeated 30 times. Results of this test shall comply with the requirements of 3.3.5.

4.6.8 Maintainability test. The comminutor shall be jammed as in 4.6.7; secure and tag power to the comminutor; drain comminutor. Manual removal of the cutter assembly, unjamming cutters, and re-assembly and restoration of the unit to operating condition shall not exceed 1 manhour as specified in 3.9.4.

4.6.9 Back-Pressure test. Simulated sewage slurry of 1 percent solids by weight shall be gravity fed through the comminutor in sufficient quantity to maintain full-pipe flow at the comminutor inlet without a down-stream flow restrictor. Sewage may be simulated using a mixture of water, and dry dog food, paper or other acceptable solid materials. The pressure drop across the comminutor under the prescribed flow condition shall not exceed 1 foot of water.

4.6.10 Jam-failure repair. Starting with the power to the comminutor secured and tagged and the unit drained, repair parts predicted and identified by 3.9.1 shall be replaced. Time to execute the simulated repair action shall not exceed 2 manhours.

4.6.11 Vibration test. The comminutor and its associated motors, controllers, and drive mechanisms shall be vibration tested under normal operating conditions as specified in 3.7.5. When specified in the contract or order, vibration testing reports shall be prepared (see 6.2.2).

4.6.12 Reliability-durability test. Test shall be run using sea water or a simulated 3-1/2 percent salt-water solution formulated from commercially available salt compounds for this express purpose. Salt water sewage shall be simulated by adding 1 percent solids, by weight, to the salt solution (see 4.6.9). The duration of the test shall be at least 250 hours and conform to the following:

(a) First test phase:

(1) Accumulates a minimum of 150 motor-controller on-off cycle. Shut comminutor off only under zero flow condition.
(2) Accumulate at least 120 hours of run time on comminutor.
(3) At least 40 hours of the 120 hours of run time shall be introduced to the comminutor intermittently at regular internals to simulate a shipboard situation where sewage will be seen in slugs. The comminutor shall be subjected to a minimum of 200 of these wet flow cycles during the 40 hours.

(b) Second test phase:

(1) Secure comminutor and leave unit installed in the dry mode for 60 hours, and the wet (submerged) mode for 60 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

(c) Third test phase:

(1) Remove cutter assembly and inspect comminutor for wear or corrosion.
(2) Reassemble comminutor and run wet for 10 hours.

Note: Other tests required by this specification may be conducted simultaneously with the reliability-durability test on a not-to-interfere basis.

4.6.12.1 Reliability-durability failure. Occurrence of any of the following during the course of the reliability-durability test shall be basis for rejection:

(a) Mechanical or electrical failure of any kind excepting failure of power supply.
(b) Maintenance or repair action.
(c) Leakage of simulated sewage or lubricant from the comminutor.
(d) Clogging of the comminutor.
(e) Indication of any deleterious corrosive or galvanic action.
(f) Indication of accelerated wear patterns which would preclude meeting service life requirements of 3,8.
(g) Excessive motor temperature rise beyond the limits specified in MIL-M-17060.

4.7 Test reports. When specified in the contract or order, test reports shall be prepared (see 6.2.2).

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation, packing, and marking. The equipment with accessories shall be preserved level A or C, packed level A, B, or C as specified (see 6.2.1), and marked in accordance with MIL-P-16789 and as specified herein.

5.2 Cushioning and wrapping materials. In addition to the requirements for cushioning materials in MIL-P-16789, the following shall apply: Use of excelsior, newspaper, shredded paper (all types, including wax paper) and similar hydroscopic or nonneutral materials and all types of loose-fill materials for applications such as cushioning, filler, stuffing, and dunnage for materials destined for shipboard stowage and use is prohibited except that vermiculite is approved for packaging applications of liquid (chemical, petroleum, and so forth) products. Cushioning and wrapping materials selected shall have properties and characteristics for resistance to fire. Examples are:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

UU-P-268 - Paper, kraft, wrapping type II, grade, C or D.
PPP-C-850 - Polystyrene, expanded grade SE, type I or II.
PPP-C-1120 - Bound fiber, uncompressed type III or IV, grade 1.
MIL-R-6130 - Cellular rubber, grade A.
MIL-R-20092 - Cellular rubber, class 5.
MIL-P-26514 - Polyurethane foam (rigid or flexible).

5.3 Talc/talcum or soapstone powder. Talc/talcum or soapstone powder used in the packaging process of items shall be free of asbestos and asbestiform-like materials.

5.4 Marking. In addition to the marking requirements specified in MIL-P-16789, the marking and labeling requirements for the schedule of quality conformance tests and visual inspection in MIL-P-116 shall apply.

6. NOTES

6.1 Intended use. The sewage comminutor and associated drive mechanism, motor, and motor controller covered by this specification are intended for Naval shipboard use in conjunction with gravity collection sewage systems. Effluent from the comminutor is discharged to holding tanks for subsequent transfer pump discharge to the open sea, barge, or shore receiving facilities. The intent of treating the sewage by comminution is twofold:

(a) Reduce the sewage solid particles sizes to a level which enables the aeration system installed in the holding tanks to maintain the solid particles in suspension. This minimizes sludge settling and build-up in the tanks.
(b) Protect the sewage transfer pumps against damage from large solid objects and particularly from long stringy materials which could foul or bind-up the impeller.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1).
(c) Quantity and size (see 3.4,4).
(d) Voltage and current if other than 440-volt, 3-phase, 60-Hz alternating current (see 3.5.2).
(e) Levels of preservation and packing required (see 5.1).
(f) When preservation, packing, and marking data is not required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no,	Option
3.16 and appendix	Engineering drawings	DI-DRPR-80651	Level 3
4.1.2	Inspection system program plan	DI-R-4803	----
4.6.5	Report, component shop noise tests	UDI-T-23764	----
4,6.6	Reports, equipment shock test	UDI-T-23753	----
4.6,11	Report, vibration testing	UDI-T-23762	---
4.7	Inspection and test reports	DI-T-5329	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L. , AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2,1 The data requirements of 6,2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical q anuals are required, Military specification and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Packaging requirements. Section 5 requires the comminutor units, associated equipment, and tools to be preserved, packed, and marked in accordance with MIL-P-16789. This specification specifically applies to the packaging of pumps, including prime movers; however, for purposes of interpreting the requirements of section 5, the “comminutor” is analogous to and interchangeable with “pump” in the referenced specification.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Cutters
Macerate
Sewage treatment system

6.8 Changes from Previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4630-N018)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is applicable only when data item description DI-DRPR-80651 is cited on the DD Form 1423,

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings, When required by the contract or order, drawings shall show outside dimensions, mounting requirements, maintenance and withdrawal space requirements , and all data necessary for installation and powering of the comminutor, motor, drive mechanism, and motor controller. Drawings for any special tools required for disassembly, maintenance, and repairs shall also be included.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24603A(SH)	SEWAGE COMMINUTOR, AND ASSOCIATED MOTOR, CONTROLLER, AND DRIVE MECHANISM FOR SHIPBOARD USE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:14Z
Check the source to verify that this is the current version before use.